Stormwater Overflow Events - STAT 171 Final Project Report

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Introduction

Urban drainage systems are implemented to manage stormwater runoff and reduce the risk of flooding. These systems are designed to collect rainfall runoff and conduct it through the drainage system. However, during periods of excess rainfall, overflow events may still occur when drainage systems reach capacity. Overflow events lead to degraded water quality and localized flooding. The likelihood of an overflow event occurring depends on several factors including amount of rainfall, sewer system design, land surface characteristics, and upstream construction. Identifying which factors are most strongly associated with overflow events can help improve draining infrastructure and guide targeted mitigation strategies. Using the available data, this study aims to answer the following research question: What are the key predictors of sewer overflow occurrence and intensity?

Data Description

The dataset contains 297 observations, with each observation representing a storm-catchment summary. The variable, overflow events, records the number of stormwater events observed during each storm. Rainfall accounts for the total rainfall during the storm in millimeters (mm). The centered rainfall variable is calculated by subtracting the sample mean rainfall from each observation. This mean-centered total occurs in both positive and negative values, representing above-average rainfall and below-average rainfall, respectively. The catchment type identifies the dominant catchment land-use type (Residential, Commercial, Industrial, or Mixed). Impervious surface is expressed as a percentage of impervious surface, land cover that does not allow water infiltration, in the catchment. One of the infrastructure-related variables is combined sewer, a binary variable used to indicate whether a catchment uses

a combined sewer system (1 = Yes, 0 = No). Maintenance overdue is another binary indicator for whether maintenance is overdue or not (1 = Yes, 0 = No). Whether there is upstream construction activity present during the storm is also represented with a binary variable (1 = Yes, 0 = No). Lastly, the green infrastructure index is a continuous measure that represents the extent or effectiveness of green stormwater infrastructure within the catchment. The higher values indicate greater implementation of features that promote infiltration and reduce runoff.

Before fitting models, we conducted exploratory data analysis (EDA) to understand the variables individually and assess their relationship with overflow counts. The distribution of overflow events was right-skewed, indicating that most storms lead to lower counts of overflow events. The mean (2.616) and variance (4.393) suggests overdispersion because the variance of overflow events exceeds the mean.

Because our response variable, overflow events, is a count variable, we will focus on three primary components of our EDA: the distribution of key variables, the relationship between the numeric variables, and the comparisons across categorical and binary variables. For the numeric variables, we computed descriptive statistics, and generated pairwise plots to assess distribution shape and linear association patterns. Based on these results, rainfall showed the clearest positive relationship with overflow counts ($r = 0.349$), impervious surface percentage showed a weaker but still positive relationship ($r = 0.227$), and green infrastructure showed essentially no linear relationship with overflow counts ($r = 0.018$).

Then, we compared average overflow counts across the categorical and binary predictors. Catchments with combined sewer systems had a higher mean overflow count (3.11) than catchments without combined sewers (2.18). Similarly, catchments with overdue maintenance had a higher mean overflow count (3.21) than those without overdue maintenance (2.30). We

also examined mean overflow counts by catchment type, but those differences were not as compelling as the binary infrastructure-related variables.

We used histograms, violin plots, bar plots, and faceted scatterplots for a visual check on group differences and possible interaction patterns before those terms were formally modeled. The rainfall-versus-overflow plot colored by sewer type suggested that rainfall and sewer design might jointly affect overflow intensity, which motivated us to test interaction terms later in the analysis.

Methodology

We began the process by importing the stormwater overflow dataset and cleaning it before proceeding with modeling. Rows with missing values were removed using drop_na(), duplicate rows were removed using distinct(), and the analysis was restricted to the variables relevant to the research question: overflow_events, rain_mm, impervious_pct, combined_sewer, maintenance_overdue, upstream_construction, catchment_type, and green_infra_index. The response variable, overflow_events, is a count variable, so we first checked whether a Poisson-type framework might be appropriate. Summary statistics showed that overflow_events had a mean of 2.616 and a variance of 4.393, with values ranging from 0 to 13. Since the variance exceeded the mean, the response exhibited overdispersion, which made standard Poisson regression less appropriate. For that reason, we proceeded with a negative binomial regression with a log link selected as the main modeling approach, since it is designed for count data with extra-Poisson variation.

After EDA, we assessed multicollinearity by using VIF scores. All VIF values were close to 1, with adjusted GVIF values near 1 for all predictors and the catchment type term also

showing a very small adjusted value. Since all VIF values were well below common concern thresholds, we concluded that multicollinearity was not a problem, and the predictors could be included in the same model without concern of coefficient estimates being distorted by redundancy among the explanatory variables.

The first fitted model we used was a first-order negative binomial regression including all main effects: overflow_events ~ rain_mm + impervious_pct + combined_sewer + maintenance_overdue + upstream_construction + catchment_type + green_infra_index. This model was used as the starting point because it addressed the research question directly while allowing each predictor to be evaluated after adjusting for the others. In this model, rain_mm, impervious_pct, combined_sewer, and maintenance_overdue were all statistically significant, while upstream_construction, catchment_type, and green_infra_index were not. The full model had null deviance 310.85, residual deviance 220.27, AIC 777.17, estimated dispersion parameter theta 17.19, and a global Wald chi-square statistic of 87.6 on 9 degrees of freedom with p-value approximately 5 × 10^(-15), indicating that the predictors as a set contributed significantly towards explaining variation in overflow counts.

Because the EDA suggested the possibility of more complex relationships, we further tested two expanded models with interactions. The first added interaction terms were between rainfall and imperviousness and between rainfall and sewer type: overflow_events ~ rain_mm*impervious_pct + rain_mm*combined_sewer + maintenance_overdue + upstream_construction + catchment_type + green_infra_index. The second added both those interaction terms and a quadratic rainfall term through poly(rain_mm, 2). We took these models into consideration to check whether the effect of rainfall depended on land cover or sewer

design, and whether the rainfall-overflow relationship showed curvature rather than simple linearity on the log scale.

We conducted Likelihood ratio tests to formally compare the more complex models to the first-order model. We also tested a smaller interaction-only model: overflow_events ~ rain_mm*impervious_pct + rain_mm*combined_sewer. This model was mainly used as a comparison to check whether interaction structure alone could explain the data. Then, we fit a reduced main-effects model containing only rain_mm, impervious_pct, and combined_sewer. However, since the EDA determined that maintenance_overdue was also a significant variable, we ultimately finalized the following model: overflow_events ~ rain_mm + impervious_pct + combined_sewer + maintenance_overdue.

We reviewed model diagnostics for each negative binomial fit using the standard four diagnostic plots: Residuals versus Fitted, Normal Q-Q, Scale-Location, and Residuals versus Leverage. We used these figures to check whether there were any major systematic patterns in the residuals, whether variance appeared to change sharply with fitted values, and whether any observations had unusually high leverage. To assess out-of-sample performance, we ended off the analysis by fitting the final model structure to testing data.

Results

AIC Model Evaluation:	DF	AIC Value
Full Model	11	777.1676
Full Model w/ Interaction Terms	13	779.1564
Full Model w/ Quadratic & Interaction Terms	14	780.6336
Interaction Terms Only	7	779.4635
Reduced Model	5	777.9010
Final Model	6	771.6579

Likelihood Ratio Tests	
Full vs Interactions	0.3658393
Full vs Quadratic + Interactions	0.4691799
Full vs Interactions Only	0.0357273
Full vs Reduced	0.04746966
Interactions Only vs Reduced	0.2955961
Final vs Reduced	0.004090648

Figure 1: Table on the left showing the degrees of freedom and AIC value for each of the tested models. Table on the right showing the p-values for the corresponding Chi-Square value for each tested model as a result of the Likelihood Ratio Test.

As a result of testing the model with interaction terms and the model with the quadratic + interaction terms, we observed that neither model improved the overall fit. The interaction model had a higher AIC (779.16) compared to the full model with all of the terms (777.17), and the likelihood ratio tests (LRT) indicated that the additional terms did not significantly improve the model ($p = 0.3658$). Similarly, the quadratic + interaction model produced an even higher AIC (780.63) and was supported by the LRT ($p = 0.4692$).

We then evaluated the model with the interaction terms only. Although we observed signs of significance for the first time with the LRT p-value being 0.0357 (<0.05), the AIC value of the interaction term only model (779.46) ended up being higher than the full model. As a result, we proceeded with further testing.

Then a reduced model including only rain_mm, impervious_pct, and combined_sewer was tested. Once again, the LRT produced significant results, with the p-value being 0.0474

(<0.05). Even though the lowest AIC value (777.9) had been produced thus far, it was still slightly higher than the full model AIC. It was clear that a reduced model was in the right direction of getting to a model with statistically significant results, but an alternative reduced model had to be tested. We did this by including a variable we deemed significant from our EDA.

The negative binomial regression model we ultimately finalized was: overflow_events ~ rain_mm + impervious_pct + combined_sewer + maintenance_overdue. Maintenance_overdue was not originally included in our initial reduced model, but the following AIC and LRT results prove that it was essential to include. This model produced the lowest AIC value (771.66) and the most significant p-value (0.004) from the LRT as opposed to all the other models we tested, and made it the preferred model in terms of parsimony and interpretability.

	Estimate (β)	**Standard Error**	**P-Value**
(Intercept)	-0.48008599	0.180368849	7.774941e-03
rain_mm	0.01994522	0.003008724	3.376676e-11
impervious_pct	0.01182833	0.002910396	4.820456e-05
combined_sewer	0.34965152	0.095937265	2.678235e-04
maintenance_overdue	0.28357420	0.095959197	3.125131e-03

Figure 2: A summary of regression results for overflow events for the Final model.

When looking at the p-values for the predictors, all four of them are observed to be the most valuable predictors to determine stormwater overflow occurrence. For every additional millimeter of rain, the expected number of overflow events increases by about 2% ($e^{\beta 1}$). For

every 1% increase in impervious area, the expected overflow count increases by about 1.19% ($e^{\beta 2}$). Catchments with combined sewer systems are expected to have 41.86% ($e^{\beta 3}$) more overflow events than those with separate systems, assuming rainfall and maintenance status are the same. When maintenance is overdue, the expected number of overflow events rises by 32.79% ($e^{\beta 4}$) compared to catchments where maintenance is up to date.

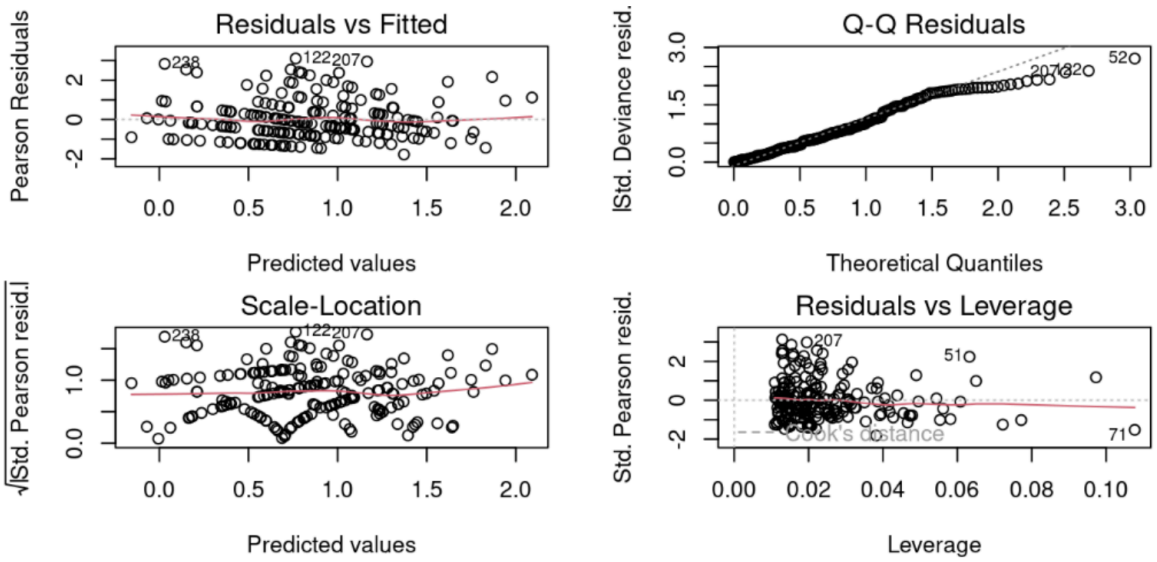


Figure 3: Standardized diagnostic plots (Residuals vs. Fitted, Normal Q-Q, Scale-Location, and Residuals vs. Leverage) of the finalized model for residual analysis to check the model's assumptions.

The diagnostic plots for the final model indicate that the Negative Binomial assumptions are reasonably met. The Residuals vs Fitted plot shows no strong non-linear patterns, though it displays the typical banding associated with count data, The Normal Q-Q plot shows some slight deviation in the upper tail, showcasing that a few outliers are present, but the Residuals vs Leverage plot confirms that no individual observations have a strong influence on the model coefficients (Cook's distance < 0.5).

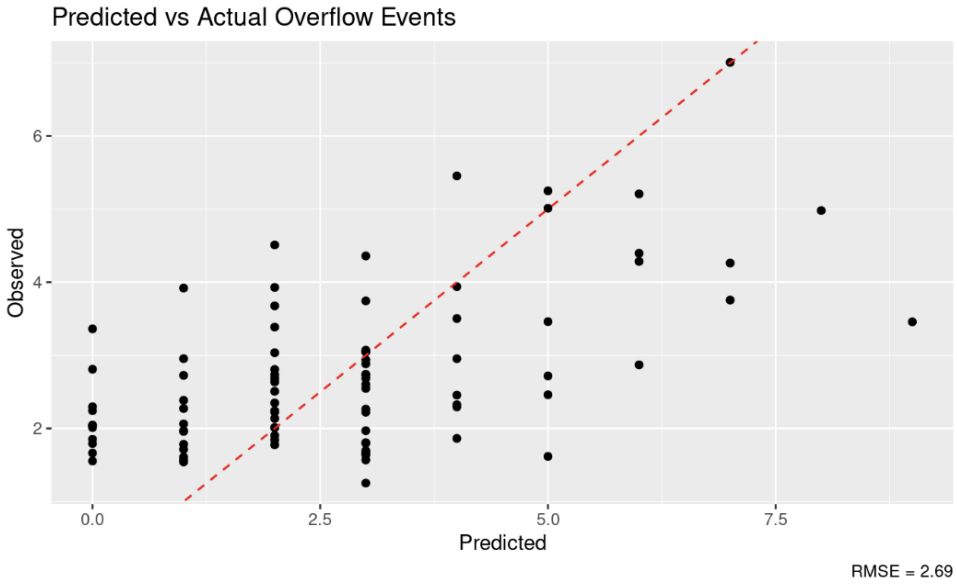


Figure 4: Scatterplot of observed overflow counts plotted against the final model's predictions for the testing data, with RMSE assessment.

When the model was run on the testing set, there was a convergence error. However, a negative binomial model was still the correct decision, as the model converged when fitted to training data. Since the testing data sample was small and only slight overdispersion was present, this error occurred. Rainfall, impervious percentage, and combined sewer remained statistically significant, while maintenance overdue was no longer significant at the 0.05 level. The testing data also showed a lower dispersion ratio than the training data, with values of 1.4137 and 1.8108, respectively. We evaluated predictive performance using RMSE, which was 2.685. In addition, we used a predicted vs actual scatterplot with a 45-degree reference to help visually assess model performance. The plot showed that the predictions followed the overall upward trend in the observed overflow counts reasonably well, although there was some variability around the reference line, indicating that the model captured the general pattern better than the exact value of every individual observation.

Discussion & Conclusion

Overall, rain_mm, impervious_pct, combined_sewer, and maintenance_overdue appear to be the most valuable predictors to determine stormwater overflow occurrence. They all have positive relationships with the response variable, meaning that if rain increases, land is more impervious, catchments have more combined sewers, and maintenance is overdue, more overflow events will occur. Therefore, to mitigate these occurrences, sewers should not be combined and maintenance must be done on time. Rainfall and land imperviousness cannot be controlled, but should be noted and monitored to determine proactive solutions to minimize overflows.

The model minimizes error as much as possible, with residual mean squared error being around 2.6. Since the range of overflow events is 13, an error of 2-3 events is negligible. However, some key limitations must be noted. The model is trained and tested on a limited sample, and additional data would greatly improve its accuracy. Moreover, the location of the urban drainage systems this data was collected from is unknown. Therefore, the results of this model cannot be applied to all systems, as these same factors may have different effects on overflows in various other parts of the world.

Next steps would be to gather further data, along with more information like the location and how these systems were constructed. The model should be trained on these new samples, with new predictors being tested to see if they improve accuracy. The data samples should be larger than the current dataset, so that the convergence error does not occur when the training/testing split is implemented.

Author Contributions

Megan Nakajima wrote the introduction.

Sumedha Chalasani & Megan Nakajima performed EDAs.

Megan Nakajima wrote the data descriptions.

Megan Nakajima & Swaraj Dash analyzed and wrote the results from the EDA.

Rhea Verma outlined the methodology/analysis plan.

Swaraj Dash, Rhea Verma, and Sumedha Chalasani performed the model fitting/evaluation.

Swaraj Dash wrote the methodology section.

Rhea Verma wrote the results section.

Sumedha Chalasani wrote the discussion & conclusion section.

Sumedha Chalasani & Rhea Verma formatted the rmd file and the written report.

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